

Education

- Expected May 2028** **Ph.D. in Electrical Engineering (Medical Imaging & AI)**, *Georgia Institute of Technology*
 Graduate GPA: 3.9/4.0
 Research focus: machine learning and representation learning methods for medical imaging analysis, with experience in large-scale brain MRI datasets and diverse downstream tasks including tumor segmentation, survival prediction, neurological disease diagnosis, and cross-cohort generalization.
- May 2023** **B.S. in Electrical Engineering (2nd Major: Computer Science)**, *University of Wisconsin–Madison*
 Cumulative GPA: 3.7/4.0 | EE Major GPA: 3.9/4.0
 Dean's List.

Research Projects

BrainDINO: Self-supervised Foundation Model for Brain MRI Aug 2025 – Present

- Developed a large-scale self-supervised representation learning framework for brain MRI using **52k+ 3D volumes** aggregated from heterogeneous public neuroimaging datasets.
 - Designed a DINO-style teacher–student training strategy for slice-wise MRI representation learning with strict subject-level separation to study robustness under cross-dataset distribution shift.
 - Evaluated representation transfer using frozen-backbone adaptation under **limited-label regimes** (10–100% supervision), across diverse clinical tasks including tumor segmentation, disease classification (autism spectrum disorder, mild cognitive impairment), survival prediction, and molecular mutation prediction.
- Outcome: First-author manuscript in preparation.*

MedDINOv3: Domain-adaptive Vision Foundation Model Jun 2025 – Oct 2025

- Studied domain-adaptive self-supervised pretraining for dense visual representation learning under large appearance and distribution shift.
- Adapted the DINOv3 self-distillation framework and refined a ViT-based architecture with multi-scale token aggregation for large-scale medical imaging datasets.
- Evaluated cross-modality and cross-domain transfer between CT and MRI benchmarks using the **CT-3M dataset** (3.8M+ slices across 16 datasets). *Outcome: arXiv preprint.*

Brain Metastases Detection and Segmentation Dec 2024 – May 2025

- Developed a **3D deformable convolutional neural network** for detecting and segmenting small brain metastases in MRI, improving robustness to scale variation and sparse lesion distribution.
- Outcome: First-author paper in Medical Physics (2025).*

Reconstruction-guided Self-supervised Segmentation (AnatoMask) 2023 – 2024

- Proposed reconstruction-guided self-masking strategies for **self-supervised segmentation**.
- Improved robustness and generalization under **limited-label supervision** through reconstruction-guided representation learning.
- Outcome: ECCV 2024 paper.*

Publications

- [1] **Wu, Y.**, Li, Y., Hu, M., Chang, C.-W., Qiu, R. L. J., Wang, T., Shu, H.-K., Mao, H., Tian, Z., & Yang, X. (2025). *Detection and Segmentation of Brain Metastases on MRI Using 3D-MedDCNet*. *Medical Physics*, 52(7), e18001.
- [2] Li, Y., Luan, T., **Wu, Y.**, Pan, S., Chen, Y., & Yang, X. (2024). *AnatoMask: Enhancing Medical Image Segmentation with Reconstruction-guided Self-masking*. *European Conference on Computer Vision (ECCV)*, 146–163.
- [3] Li, Y., **Wu, Y.**, Lai, Y., Hu, M., & Yang, X. (2025). *MedDINOv3: How to Adapt Vision Foundation Models for Medical Image Segmentation?* *arXiv preprint arXiv:2509.02379*.
- [4] Li, Y., Wynne, J. F., **Wu, Y.**, Qiu, R. L. J., Tian, S., Wang, T., Patel, P. R., Yu, D. S., & Yang, X. (2025). *Automatic Medical Imaging Segmentation via Self-supervising Large-scale Convolutional Neural Networks*. *Radiotherapy and Oncology*, 204, 110711.
- [5] Zhao, F., **Wu, Y.**, Hu, M., Chang, C.-W., Liu, R., Qiu, R., & Yang, X. (2025). *Current Progress of Digital Twin Construction Using Medical Imaging*. *Journal of Applied Clinical Medical Physics*, 26(9), e70226.

Professional Service

Reviewer for *IEEE Transactions on Medical Imaging* (2025); *Medical Physics* (2026); *MICCAI* (2025)

Industry Experience

MRI Scout-based Anatomical Detection for Safety-Critical Systems May 2024 – Aug 2024, *GE Healthcare*
MR Application / PSD Development Engineer Intern

Developed safety-critical 3D computer vision models for robust anatomical detection under partial observability and truncated fields of view; queried internal clinical imaging databases using **SQL** to identify and retrieve MRI cohorts for model development and validation; 3D MRI SAR GradWarp correction for reconstruction; system-level implementation in **C++**; built end-to-end data pipelines including data collection, labeling, training, evaluation, and deployment; containerized deployment using **Docker** with reproducible ML workflows under strict reliability constraints.

Outcome: Deployed as an internal safety modeling component at GE Healthcare.

Technical Skills

Programming	Python, C/C++, MATLAB, SQL
Deep Learning	PyTorch, TensorFlow; Vision Transformers (ViT), convolutional neural networks (CNN); self-supervised learning, contrastive learning, masked modeling, self-distillation, transfer learning
Evaluation & Analysis	representation evaluation (CKA), embedding visualization (t-SNE), robustness analysis, statistical validation
Medical Imaging Systems	multimodal medical imaging analysis, survival modeling (Cox regression), cross-domain transfer Linux/Unix, Git, Docker, reproducible ML pipelines

Teaching Experience

Feb 2022 – May 2023 **Undergraduate Teaching Assistant**, *University of Wisconsin–Madison, College of Engineering*
 ECE 532: Matrix Methods in Machine Learning
 ECE 252: Introduction to Computer Engineering
 Led weekly office hours and discussion sessions for classes of up to 50 students. Provided technical guidance on linear algebra, machine learning fundamentals, and programming assignments. Assisted with grading, debugging support, and student mentoring throughout the semester.